

Implied Volatility Surface Construction

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2Q 2019

Version Control

Version	Date	Authors	Comments
1.0	5/16/2019	Manu Maurette	First draft.
1.1	5/28/2019	Manu Maurette, Emma Li	First version

1 Introduction

The goal of this document is to study the possibility of generating in house volatility surfaces. Currently Axioma uses a Vendor to obtain Volatility Surfaces. These are then used as inputs in AxR pricing models.

2 Input Data

We here give a brief description of the necessary inputs for obtaining implied volatilities

2.1 Option Prices

For a given security (Stock, Index), for each listed Strike, Expiration and Option Type (Call/Put), the market price is needed. This information in general is provided by data vendors as:

- Best Bid price available
- Best Ask price available
- Last traded date
- Daily Volume
- Open interest

The first two are essential. The last three, instead, **could be used for cleaning the data and avoiding using illiquid products that might affect the calculations.**

Currently AXIOMA obtains that data from Thomson Reuters in *EJV_derivs* database (available in Pandora, for example)

We will follow the following assumptions:

- Market price of an option to be $\frac{Bid+Ask}{2}$ (industry standard)
- No lag for settlement needs to be taken into account.
- No filling for missing quotes of option prices.

2.2 Risk Free rates

Pricing models also assume the existence of a risk-free rate used for discounting. We will assume this curve is the Zero Coupon USD (or the corresponding currency) Swap Rate curve.

Currently, AXIOMA obtains these rates from Thomson Reuters in *MarketData* database (available in Pandora, for example). See Testing section for an example.

In addition, the rates obtained from the database have to be interpolated to the options times to maturities. This will be continued in the Analytics section

2.3 Dividend Yield

2.3.1 Equity

Currently, AXIOMA obtains these rates from the *Oracle Equity* database. See Testing section for an example.

2.3.2 Indexes

In the case of European, Canadian and Asian Indexes, an industry standard is to estimate the dividend as $q = \frac{PV(div)}{T \cdot S}$.

We are still in review on alternative ways to obtain these

3 Modeling

There are several pieces in the process that require modeling decision:

3.1 Interest Rate curve interpolation

As mentioned in the previous section, rates have to be interpolated to get rates for the listed options times to maturities. This can be done with several methods as:

- Piecewise Constant
- Piecewise Linear
- Cubic Spline

3.2 Dividend Yield

Estimation of the dividend yield also require a modeling decision.

There are several methods to estimate an Index Dividend Yield. For the US case, an industry standard is to use a Call-Put parity based regression. For dividend paying stock, a call put parity relationship holds:

$$S_0 - PV(Div) + P = C + Ke^{-rT},$$

so, by approximating $PV(Div) = S_0(1 - e^{-dT})$

$$C(t) - P(t) = S_0e^{-dT} - Ke^{-rT}.$$

By first order approximation of $e^{-rT} = 1 - rT$:

$$C(t) - P(t) = S_0(1 - dT) - K(1 - rT),$$

regrouping:

$$C(t) - P(t) = S_0 - dS_0T - K - rKT.$$

The following regression is done:

$$C - P = b_0 + b_1S + b_2ST + b_3K + b_4KT + b_5D_{BA}.$$

The regression is difference between the price of a call option and the price of a put option with the same expiration and strike. When calculating this difference, the bid price of the call is used with the offer price of the put, and vice versa. D_{BA} is a dummy variable set equal to 1 if the call option's bid price is used. S is the underlying security's (index) closing price, K is the strike price of the call and put options, and T is the time to expiration in years. The regression is calculated using three months of option data across all strikes and expirations with an exception of

contracts expiring in less than 15 days, for a single underlying. According to the principle of put-call parity, the dividend yield on the underlying index will be approximately equal to the negative of the estimated parameter b_2 .

3.3 Option Analytics

3.3.1 American

Most US stock options have American style exercise. This requires the need of an American Option model implementation. Some common choices are:

- CRR binomial tree model
- Modification of binomial tree models
- LSMC method
- Axioma pricers (may include the previous)

3.3.2 European

For European style options (most US indices are of European nature) the natural choice is the Black Scholes model.

3.4 Root Finding algorithm for obtaining implied volatilities

Again, this step requires a modeling decision:

- Bisection
- Newton Raphson
- Jackel implication (**cite lets be rational**) for European style.

3.5 Volatility Surface interpolation/extrapolation

Once all the implied volatilities, we need to generate a standard volatility surface. Again, this is an interesting mathematical problem. Some algorithms for doing this include:

- Piecewise linear
- Cubic spline
- SVI-BDK (available in AxR)
- Tension spline / Thin plate spline
- Gaussian kernel smoothing (Vega weighted)

3.5.1 Gaussian Kernel Smoothing

We generate a *standard* volatility surface based on Delta and Time to Maturities. We define standard deltas and standard TTMs:

- TTMs [30, 60, 91, 122, 152, 182, 273, 365, 547, 730] (in days)
- Deltas $\pm[20, 25, 30, 35, 40, 45, 50, 55, 60, 65, 70, 75, 80]$ (in %)
- Call-Put

For each grid point j (Delta, Time to Maturity, Type), which we perform a smoothing algorithm:

$$\hat{\sigma}_j = \frac{\sum_i V_i \sigma_i \Phi(x_{ij}, y_{ij}, z_{ij})}{\sum_i V_i \Phi(x_{ij}, y_{ij}, z_{ij})}$$

where i is indexed over all listed options for that day (after the data cleanse); V_i is the option vega, σ_i is the implied volatility and $\Phi()$ is a kernel function:

$$\Phi(x, y, z) = \frac{1}{\sqrt{2\pi}} e^{-\left[\frac{x^2}{2h_1} + \frac{y^2}{2h_2} + \frac{z^2}{2h_3}\right]}.$$

with:

- $x_{ij} = \log(T_i/T_j)$
- $y_{ij} = \Delta_i - \Delta_j$
- $z_{ij} = I_{\{CP_i=CP_j\}}$

The kernel bandwidth parameters are chosen empirically: $h_1 = 0.05$, $h_2 = 0.005$, $h_3 = 0.001$.

4 Testing and Validation

- Replicate implied volatilities for equity index: SPX, DJIA, NASDAQ, FTSE, Russell, Nikkei
- Implication only
- Full replication
- Replicate implied vols for single-name stock
- Implication only
- Full replication

4.1 First set of tests

4.1.1 AAPL stock for 04.04.2019

The following describes an end to end tests to generate volatility surface for AAPL, for 04.04.2019.

EJV Option data

In SQL, we perform the following Query

```
Declare @AsOfDate Date='2019-04-04',
@UndlyRic varchar(100) = 'AAPL.O'
if object_id('tempdb..#EJVDerivsData', 'U') is not null drop table #EJVDerivsData
select bid_px, ask_px, universal_bid_px, universal_ask_px, vol, open_interest, a.ric_root,
unscaled_strike_px, put_call_indicator, expiration_dt, exercise_style_cd, a.ric
into #EJVDerivsData
from EJV_Derivs.dbo.quote_xref a, EJV_Derivs.dbo.price b
where underlying_ric=@UndlyRic and put_call_indicator is not null and
expiration_dt >= @AsOfDate and a.quote_id=b.quote_id and b.trading_dt=@asOfDate
select * from #EJVDerivsData b
```

	bid_px	ask_px	universal_bid_px	universal_ask_px	vol	open_interest	ric_root	unscaled_strike_px	put_call_indicator	expiration_dt	exercise_style_cd	ric
1	113.75	117.8	113.75	117.8	-9999401	-9999401	AAPL.U	80	C	2019-06-21 00:00:00.000	A	AAPLF211908000.U
2	0.06	0.25	0.06	0.25	3	-9999401	AAPL.U	75	P	2020-06-19 00:00:00.000	A	AAPLR192007500.U
3	3.2	5.9	3.2	5.9	-9999401	-9999401	AAPL.U	290	C	2021-06-18 00:00:00.000	A	AAPLF182129000.U
4	65.05	68.1	65.05	68.1	3	-9999401	AAPL.U	135	C	2021-06-18 00:00:00.000	A	AAPLF182113500.U
5	10.25	12	10.25	12	-9999401	-9999401	AAPL.U	155	P	2021-06-18 00:00:00.000	A	AAPLR182115500.U
6	20.5	22.65	20.5	22.65	-9999401	-9999401	AAPL.U	185	P	2021-06-18 00:00:00.000	A	AAPLR182118500.U
7	12.1	15.95	12.1	15.95	-9999401	-9999401	AAPL.U	240	C	2021-06-18 00:00:00.000	A	AAPLF182124000.U
8	37.2	41.6	37.2	41.6	2	-9999401	AAPL.U	220	P	2021-06-18 00:00:00.000	A	AAPLR182122000.U
9	31.4	31.75	31.4	31.75	11	-9999401	AAPL.U	165	C	2019-05-17 00:00:00.000	A	AAPLE171916500.U
10	4.05	4.15	4.05	4.15	20	-9999401	AAPL.U	170	P	2019-10-18 00:00:00.000	A	AAPLV181917000.U
11	99.05	102.8	99.05	102.8	-9999401	-9999401	AAPL.U	95	C	2019-10-18 00:00:00.000	A	AAPLJ181909500.U
12	0.01	0.02	0.01	0.02	24	-9999401	AAPL.U	255	C	2019-05-17 00:00:00.000	A	AAPLE171925500.U
13	1.98	2	1.98	2	1877	-9999401	AAPL.U	210	C	2019-05-17 00:00:00.000	A	AAPLE171921000.U
14	-9999401	0.01	0.01	0.01	10	-9999401	AAPL.U	265	C	2019-05-17 00:00:00.000	A	AAPLE171926500.U
15	17.65	17.85	17.65	17.85	1	-9999401	AAPL.U	205	P	2019-10-18 00:00:00.000	A	AAPLV181920500.U
16	45.85	46.65	45.85	46.65	3	-9999401	AAPL.U	150	C	2019-05-17 00:00:00.000	A	AAPLE171915000.U
17	23.1	23.35	23.1	23.35	14	-9999401	AAPL.U	180	C	2019-10-18 00:00:00.000	A	AAPLJ181918000.U
18	15.6	14.65	15.6	14.65	374	-9999401	AAPL.U	145	C	2019-05-17 00:00:00.000	A	AAPLE171914500.U

Figure 1: EJV derivs query for Option prices

We compared option prices (bids and asks) from EJV derivs database and The Vendor and reached to the following conclusion:

- The same 1604 options were listed in both databases.
- Both databases have a similar behavior for illiquid options. One (Vendor) puts 0 in the Bid, the alternative puts 0.05 (or 0.01)
- Removing the above cases (1422 cases), 97% of the relative Bid differences fall in the (-5%, 10%) column of the histogram (see Figure below).
- Also, removing the above cases (1422 cases), 89% of the relative Ask differences fall in the (-8%, 2%) column of the histogram (see Figure below).

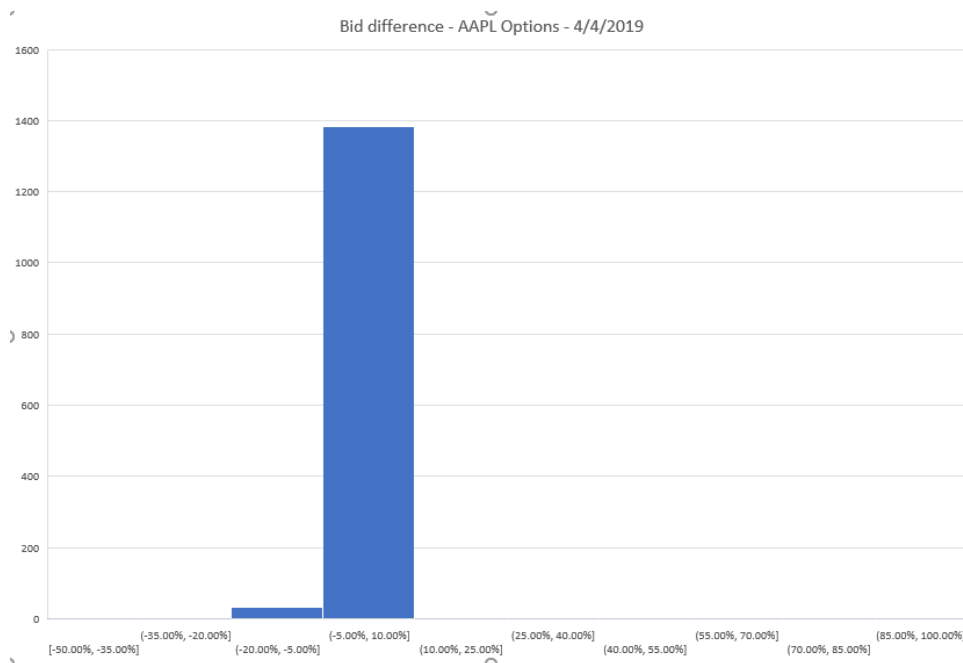


Figure 2: Bid relative difference histogram (in %)

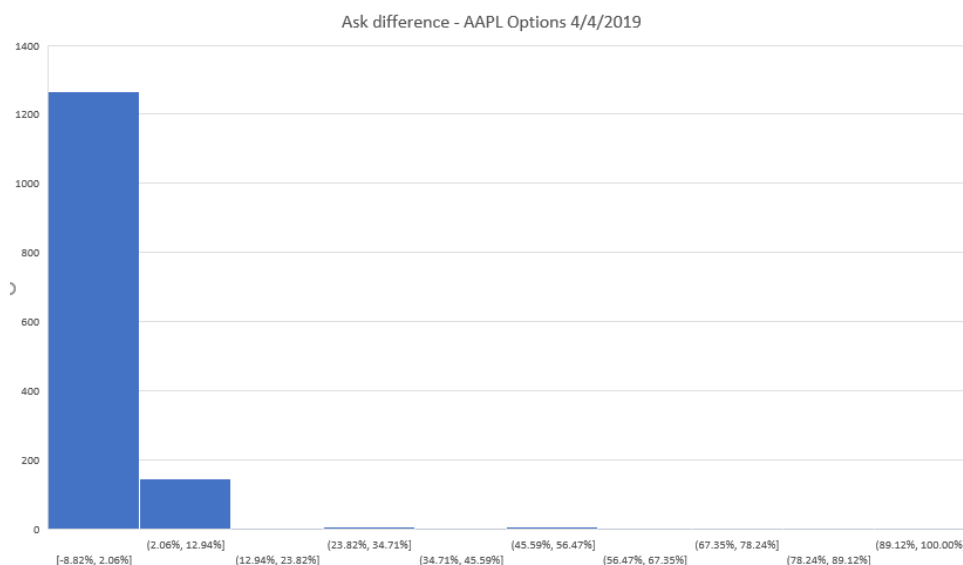


Figure 3: Ask relative difference histogram (in %)

ZC Swap Curve data In SQL, we perform the following Query

```
SELECT cv.CurrencyEnum, CountryEnum, cv.CurveTypeEnum, CurveShortName,
CurveLongName, TradeDate, te.Inyears, te.Lud, te.PeriodUnitEnum, te.Description,
te.Lub, te.NumberOfUnit, cq.Quote
FROM [MarketData].[dbo].[Curve] cv
join [MarketData].[dbo].[CurveNodes] cn on cv.CurveId = cn.CurveId
join [MarketData].[dbo].[TenorEnum] te on cn.TenorEnum = te.TenorEnum
```



```
join [MarketData].[dbo].[CurveNodeQuote] cq on cq.CurveNodeId = cn.CurveNodeId
and TradeDate = '2019-04-04'
and cv.CurrencyEnum = 'USD'
and cv.CurveTypeEnum in ('SwapZC')
order by InYears
```

	CurrencyEnum	CountryEnum	CurveTypeEnum	CurveShortName	CurveLongName	TradeDate	InYears	Lud	PeriodUnitEnum	Description	Lub	NumberOfUnit	Quote
1	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	0.002739726	2011-08-15 14:59:55.987	D	1 Day	Izaks	1	0.02544774
2	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	0.01369863	2011-08-15 14:59:55.987	D	5 Days	Izaks	5	0.02493583
3	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	0.038461538	2012-11-27 15:39:42.207	W	2 Week	avaysbrod	2	0.0246838
4	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	0.057692308	2012-11-27 15:39:42.207	W	3 Week	avaysbrod	3	0.02481099
5	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	0.0833333333333	2011-08-23 15:18:09.067	M	1 Month	pijacob	1	0.02498057
6	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	0.166666667	2011-08-15 14:59:55.987	M	2 Months	Izaks	2	0.02565059
7	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	0.25	2011-08-15 14:59:55.987	M	3 Months	Izaks	3	0.02600842
8	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	0.5	2011-08-15 14:59:55.987	M	6 Months	Izaks	6	0.02585726
9	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	1	2011-08-15 14:59:55.987	Y	1 Year	Izaks	1	0.02556519
10	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	2	2011-08-15 14:59:55.987	Y	2 Years	Izaks	2	0.02427515
11	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	3	2011-08-15 14:59:55.987	Y	3 Years	Izaks	3	0.02355078
12	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	4	2011-08-15 14:59:55.987	Y	4 Years	Izaks	4	0.02335344
13	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	5	2011-08-15 14:59:55.987	Y	5 Years	Izaks	5	0.02344301
14	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	6	2011-12-23 11:37:51.570	Y	6 Years	pijacob	6	0.0236828
15	USD	US	SwapZC	US USD.IRSWP.ZC	USD Interest Rate Swap Curve	2019-04-04	7	2011-08-15 14:59:55.987	Y	7 Years	Izaks	7	0.02395632

Figure 4: MarketData query for SwapZC curve

After a cubic spline interpolation of these data, both from the Vendor and the Alternative Axioma sources, we obtain the following plot of the curves:

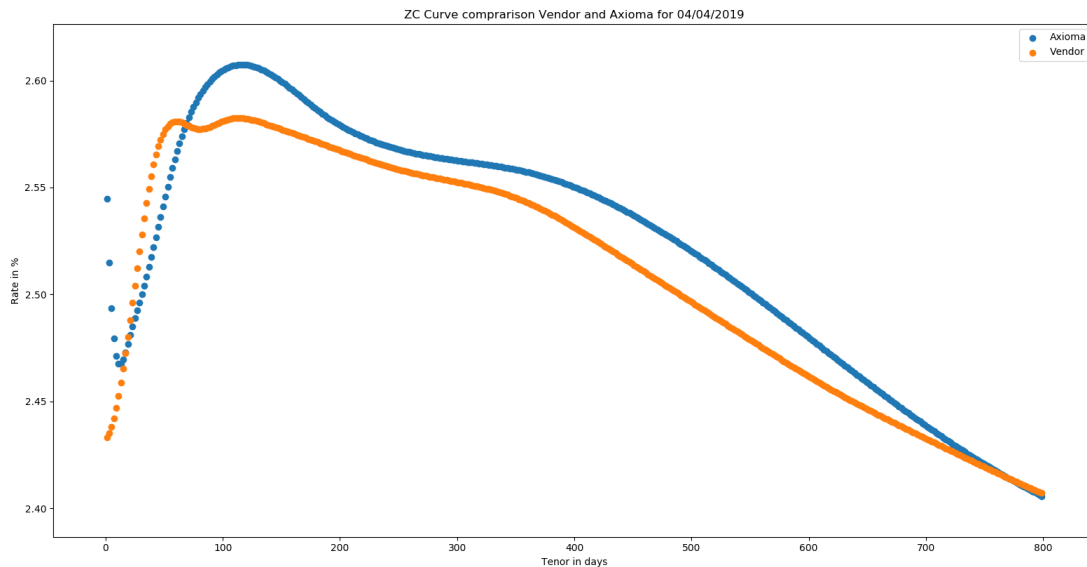


Figure 5: Differences in ZC Swap curves

There is a discrepancy in the first tenors that we currently investigating.

Dividend In SQL, we perform the following Query:

```
select * from openquery([ORA-GLPROD-marketdb_global], 'select div.dt DivDate, div.value
from modeldb_global.sub_issue_divyield_active div
where div.sub_issue_id = ''DQ8L3JXGU111''
and div.dt = TO_DATE(''2019-04-04'', ''YYYY-MM-DD'')')
```

	DIVDATE	VALUE
1	2019-04-04 00:00:00.0000000	.0151941995238886

Figure 6: Oracle Equity query for Dividend yield

A comparison was made to how the Vendor computes the dividend. It is not clear how the calculation is done, we can interpret two ways:

1. $\text{div} = \frac{\text{DividedPaid}}{\text{Spot}}$ (Past)
2. $\text{div} = \frac{\text{DividedPaid}}{\text{Spot}}$ (Estimate)

	Spot	
4/4/2019	195.69	
	Div Ax	
4/4/2019	0.015194	
	Div Vendor	DIFF
Last Div * FREQ / Spot [PAST]	0.014922	1.79%
Last Div * FREQ / Spot [FORECAST]	0.015126	0.45%

Figure 7: Differences in Dividend yield

Data cleanse aggregation

- Incorporate Options prices
- Remove NAN values of Options database
- Generate TTM (Time to Maturities) of Expiration dates based on valuation date and Expiration using (ACT/ACT) daycount convention.
- Generate risk free rate for each TTM from ZC rate data using a *cubic spine interpolation*
- Incorporate Dividend Yield
- Incorporate an IsIndex flag

Bellow is a picture of part of the DataFrame of cleaned data.

Spot	IsIndex	Dividend	Strike	TTM	Rate	CallPut	Bid	Ask
195.69	0	0.0151942	220	8	0.0247466	P	23.8	25.25
195.69	0	0.0151942	222.5	8	0.0247466	P	26.5	27.7
195.69	0	0.0151942	225	8	0.0247466	P	28.85	31.4
195.69	0	0.0151942	230	8	0.0247466	P	34.15	34.5
195.69	0	0.0151942	235	8	0.0247466	P	39.15	39.5
195.69	0	0.0151942	240	8	0.0247466	P	44.15	44.5
195.69	0	0.0151942	2.5	14	0.0246838	C	191.45	195.1
195.69	0	0.0151942	5	14	0.0246838	C	188.55	192.55
195.69	0	0.0151942	7.5	14	0.0246838	C	186.45	190.1
195.69	0	0.0151942	10	14	0.0246838	C	183.55	187.6
195.69	0	0.0151942	12.5	14	0.0246838	C	181.05	185.1
195.69	0	0.0151942	15	14	0.0246838	C	178.55	182.6
195.69	0	0.0151942	17.5	14	0.0246838	C	176.05	180.3
195.69	0	0.0151942	20	14	0.0246838	C	173.55	177.6
195.69	0	0.0151942	22.5	14	0.0246838	C	171.1	175.3
195.69	0	0.0151942	25	14	0.0246838	C	168.6	172.8
195.69	0	0.0151942	30	14	0.0246838	C	163.6	167.6

Figure 8: Clean data DataFrame

Pricing method Leisen Reisner tree method was used to price American Option. It is an enhancement of the CRR tree. ***CITE***

IV generation for each listed option in dataframe Implied volatilities are generated using a root finding algorithm. Starting from interval $\sigma \in [0, 5]$. In this case, *bisection*. Also, the LR algorithm is used in each iteration for pricing.

Generation of model outputs At this point, with the obtained Implied Volatility, the following outputs are obtained:

- Model Price
- Model Delta
- Model Vega

Here, an extra cleanse is made. Deltas smaller than 5% and greater than 95% in abs value are omitted due to potential issues in pricing calculations. **We need to study this deeper.**

At this point, it would be possible to do a comparison with the Vendor Database

Generation of volatility surface We implemented the Gaussian Kernel Smoothing

Tests Comparison tests were performed between the volatility surfaces obtained following these steps, and a volatility surface provided by a vendor.

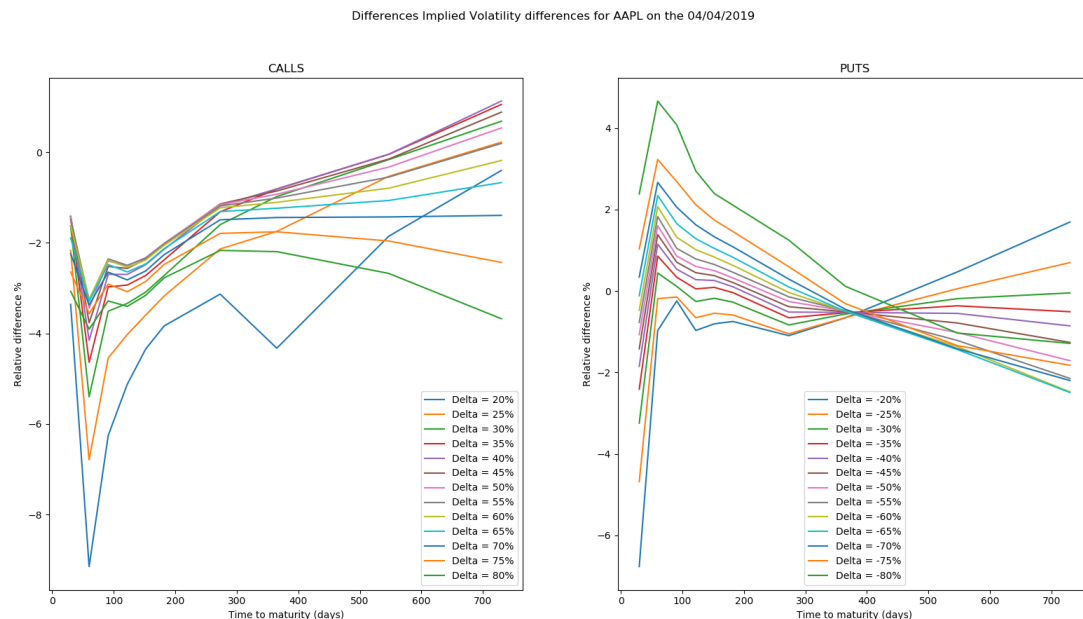


Figure 9: Differences in volatilities

We see peaks of around 9% differences for the 20% Delta (out of the money) and short maturities (1M 3M) and an average of a $\pm 2\%$ relative difference between both.

4.1.2 SPX index for 04.04.2019

The following describes an end to end tests to generate volatility surface for SPX, for 04.04.2019.

EJV Option data I was provided a Query *****FIND GENERAL QUERY***** with data of listed options.

We compared option prices (bids and asks) from EJV derivs and The Vendor and reached to the following conclusion:

From 18127* options with data both from we found:

- 1 case with ASKS relative difference 100% They correspond to Asks in Vendor1 = 0

- 26 (0.14%) cases with ASKS relative differences above (or equal) 50%: They correspond to Asks in Vendor2 in the set 0.05 and 0.1 They correspond to Asks in Vendor1 in the set 0.05, 0.1, 0.15 and 24 of them correspond to Bid in Vendor 1 = 0
- 43 (0.24%) cases with ASKS relative differences between 25% and 50%: They correspond to Asks in Vendor2 in the set 0.15, 0.2, 0.25, 0.35 They correspond to Asks in Vendor1 in the set 0.1, 0.15, 0.2, 0.25
- 183 (1.01%) cases with ASKS relative differences between 10% and 25%: All ASKS bellow 3.3 (Vendor1) and 3.7 (Vendor 2)
- 336 (10.48%) cases with ASKS relative differences between 5% and 10%: All ASKS bellow 8.1 (Vendor1) and 8.6 (Vendor 2)
- 1900 (9.9%) cases with ASKS relative differences between 1% and 5%:
- 86.27% of the cases with ASKS relative differences below 1%

ZC Swap Curve data Same as the last case, as it is the same date

Dividend We still do not have a clear idea of how to obtain Dividend for Index Options. We used 1.86% following several data sources (not from Axioma).

The vendor uses the regression algorithm described in the Analytics section.

Data cleanse aggregation

- Incorporate Options prices
- Generate TTM (Time to Maturities) of Expiration dates based on valuation date and Expiration using (ACT/ACT) daycount convention.
- Generate risk free rate for each TTM from ZC rate data using a *cubic spine interpolation*
- Incorporate Dividend Yield
- Incorporate an IsIndex flag (1 in this case)

Pricing method BSM model was used to price the European style options.

IV generation for each listed option in dataframe Implied volatilities are generated using a root finding algorithm. Starting from interval $\sigma \in [0, 8.5]$. In this case, *bisection*. Jeckel algorithm was also tested with the exact same results.

Generation of model outputs At this point, with the obtained Implied Volatility, the following outputs are obtained:

- Model Price
- Model Delta
- Model Vega

Here, an extra cleanse is made. Deltas smaller than 5% and greater than 95% in abs value ***LOOK FOR A BETTER APPROACH*** are committed due to potential issues in pricing calculations.

At this point, it would be possible to do a comparison with the Vendor Database

***INCLUDE COMPARISSON ***

Generation of volatility surface We implemented the Gaussian Kernel Smoothing

Tests Comparison tests were performed between the volatility surfaces obtained following these steps, and a volatility surface provided by a vendor:

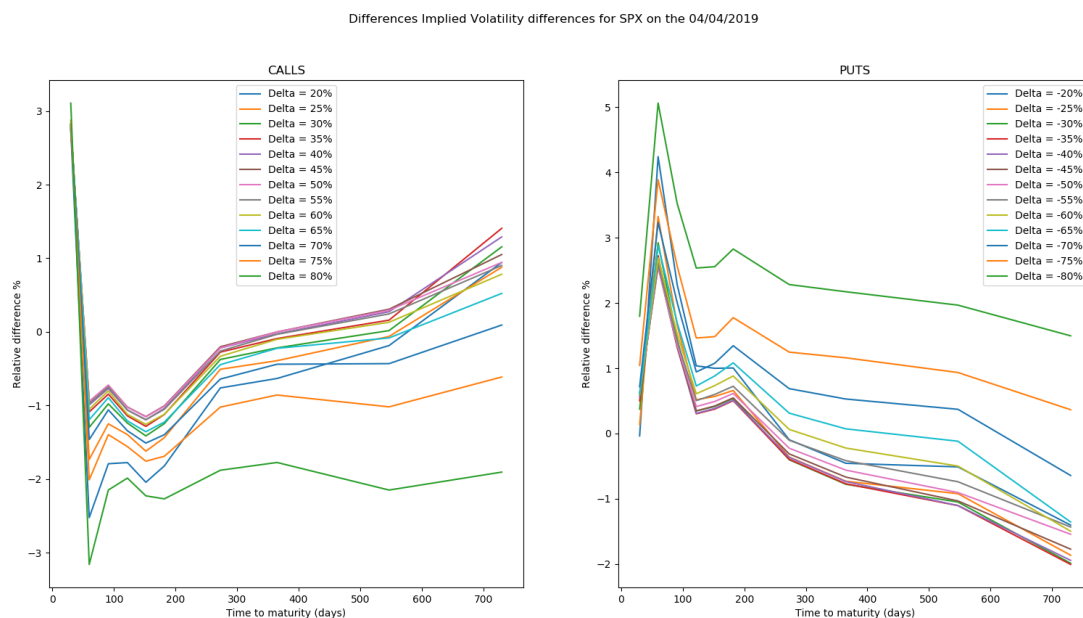


Figure 10: Differences in volatilities

We see peaks of around 3% differences for short maturities (1M 3M) and an average of a $\pm 2\%$ relative difference between both.

5 Production Work flow

- Productionizing the workflow
- What technology do we use?
- Programming language?
- Platform: Azure + Spark
- Can we compute IVS on the fly? (real-time)



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